

MD. Rafi Imam

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RESEARCH INTERESTS

Machine Learning • Natural Language Processing • Computer Vision • Multimodal Systems • AI Systems Engineering • Applied AI in Real-World Systems

EDUCATION

Rajshahi University of Engineering & Technology (RUET) 2018 – 2023

B.Sc. in Electrical & Computer Engineering CGPA: 3.30/4.00

Last 60 credit hours GPA: 3.55/4.00

Relevant Coursework: Machine Learning, Neural Network, Natural Language Processing, Image Processing, Data Structures, Algorithms, Digital Signal Processing, Computer Networks

PUBLICATIONS

1. **R. Imam**, et al., *Suicidal Thought Detection Using NLP on Reddit Data*, IEEE ICCIT 2023.
2. Md. M. Hossain, et al., **R. Imam**, *Privacy-Preserving Federated Learning for Lung Cancer Classification*, IEEE ICCIT 2023.
3. T. H. Mou, et al., **R. Imam**, *Detecting Polycystic Ovary Syndrome Using Machine Learning*, IEEE ICCIT 2023.
4. Z. Afrin, et al., **R. Imam**, *YOLOv8-based Object Detection for Self-driving Cars*, IEEE ICCIT 2023.

RESEARCH & PROFESSIONAL EXPERIENCE

Assistant Officer (Software Development) Dec 2023 – Present
Jamuna Bank PLC, ICT Division

- Designing and deploying large-scale systems in a **Linux-based production environment**, focusing on scalability, reliability, and secure architecture.
- Worked on **AI-integrated financial systems**, including OCR pipelines, face matching, and structured data extraction for real-world decision workflows.
- Developed **data processing and validation pipelines** involving AML checks, deduplication, and transaction profiling for risk-aware systems.
- Built **event-driven and distributed systems** using Kafka and workflow orchestration tools for asynchronous processing.
- Applied **real-time system design** using WebSocket and push notification services.

SELECTED SYSTEMS & PROJECTS

eKYC Platform (Digital Account Opening System)

.NET Core, React, Tailwind, Python, FastAPI

- Designed full-stack system for digital onboarding with multi-stage approval workflows.
- Integrated **AI-based verification pipeline**:
 - NID OCR and face matching (Polygon API)
 - Document OCR, address extraction, and profession prediction (Google APIs)

- Implemented AML checks, deduplication, and transaction behavior capture.
- Built real-time communication system using WebSocket and Firebase.

Bancassurance Platform

Node.js, Kafka, MongoDB, Netflix Conductor

- Designed distributed workflow system integrating multiple insurance providers.
- Implemented event-driven architecture for asynchronous processing.
- Developed fault-tolerant callback and retry mechanisms.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C/C++

ML/AI: PyTorch, TensorFlow, NLP, Computer Vision, OCR systems

Systems: Linux, Distributed Systems, Kafka, REST APIs

Backend: Node.js, FastAPI, ASP.NET Core

Databases: MongoDB, PostgreSQL

Tools: Docker, Kubernetes, Git

STANDARDIZED TEST SCORES

GRE: 304 (Quantitative: **164**, Verbal: 140, AWA: 3.0)

IELTS Academic: Overall Band **7.0** (Listening: 7.0, Reading: 7.0, Writing: 6.5, Speaking: 6.5)

HONORS & AWARDS

- RUET Merit Scholarship (Top 30 students) – 2020, 2021, 2022
- Leadership Award: Class Representative – 2020–2023